A Constructive Introduction to Quantum Field Theory

UV stability and the limit

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SISSA lectures, Lecture 6 — notes: Lecture 6, §§17–19

Where we are

Lecture 5, at every lattice N and upper scale T:

$$\|E_T\|_{L^p} + \|J_T\|_{L^p} \leq C_\gamma \, (1 + p^5 + \delta_{N,T,p}), \qquad \delta_{N,T,p} = \|T Z_T\|^2_{L^{2p}(\mathbb{P};L^2)}.$$

Three things remain:

1. the boundary term vanishes: $T Z_T \to 0$ at fixed N;
2. uniform energy bounds ⇒ compactness of processes and measures;
3. identify the limit: coefficients, noise, equations.

The subtle point

Convergence of the fields does not pass an FBSDE to the limit: the backward martingale could acquire **extra noise**. The statistical meaning of the backward variable saves us.

The boundary at infinite resolution

Lecture 6

The boundary lemma

$\bar{\Gamma}_q$ was built to converge to the bare action: $\mu_q \to \mu_N$, $\Theta_q \to 0$, corrections integrated from $q = \infty$.

Lemma. (fixed N, constants depend on N)

$$\|Y_q - \Phi_\infty\|_{L^p} + \|q\, Z_q\|_{L^p} \le C_{N,p}\, q^{-1/2}, \qquad \textit{hence} \qquad \delta_{N,T,p} \le C_{N,p}\, T^{-1}.$$

- The field at resolution q: $\Phi_q = C_q P_q = (q - q_0)\, C_q \Phi_\infty + C_q B_{q-q_0} \to \Phi_\infty$.
- $\Phi_q = Y_q + C_q D\bar{\Gamma}_q(Y_q)$, so $Y_q - \Phi_q = O(q^{-1})$ and $D\bar{\Gamma}_q(Y_q) \to D V_N(\Phi_\infty)$.
- Exact minus approximate coordinate identity:

$$q\, Z_q = q\, C_q \left[D\bar{\Gamma}_q(Y_q) - \mathbb{E}_q D V_N(\Phi_\infty) \right];$$

  compare $\mathbb{E}_q G$ with the **observed** $G_q^0 = D V_N(\Phi_q)$: $\|\mathbb{E}_q G - G\| \le 2\, \|G - G_q^0\|$.

No uniform estimate, no posterior concentration.

UV stability

Let $T \to \infty$ at fixed N (paths and ν_N do not depend on T; monotone convergence; $\delta_{N,T,p} \to 0$). **Only then** take $\sup_N$.

Theorem. (UV stability) *For $q_0 = q_0(\gamma)$ large,*

$$\sup_N \left(\|E_\infty^N\|_{L^p} + \|J_\infty^N\|_{L^p}\right) \le C_\gamma (1+p^5), \qquad \sup_N \mathbb{E}\|Z_q^N\|_2^2 \le C_\gamma q^{-5/2},$$

and $\sup_N \mathbb{E} \exp\{c\,(E_\infty^N + J_\infty^N)^{1/5}\} < \infty$.

- $C_{N,p}$ diverges with N: never take $\sup_N$ of $C_{N,p}/T$.
- The force: $F_q = \mathbb{E}_q[-D V_N(\Phi_\infty)] = K_q Z_q - D\bar{\Gamma}_q(Y_q)$, the explicit force at the approximate forecast, corrected by $K_q Z_q$.

Compactness

Lecture 6

Suprema by a maximum principle

Embed by cells $\square_x$: $\iota_N : H_N \to H = L^2(\mathbb{T}^3)$ isometric, $j_N = \iota_N^*$.

Lemma. (supremum after closure, $\varpi = \frac{3}{8}$) $\mathcal{Y}_T = \sup_q q^{-\varpi} \|Y_q\|_\infty$, $\mathcal{X}_T = \sup_q q^{-\varpi} \|X_q\|_\infty$:

$$\mathcal{Y}_T \le C\,(\mathcal{X}_T + q_0^{-\varpi} \gamma^{1/2} E_T^{1/2}), \qquad \|\mathcal{X}_\infty\|_{L^p} \le C\sqrt{p}.$$

- Remove the backward forcing: $\mathcal{H}_q = C_q F_q^Z$, $\|\mathcal{H}_q\|_\infty \lesssim q^{-1/4} \|F_q^Z\|_2 \lesssim \gamma^{1/2} E_T^{1/2}$.
- At frozen q: $K_q U + \frac{\gamma}{3}(U+d)^3 = f$. Constants are super- and subsolutions: **lattice maximum principle**.
- The weight $q^{-\varpi}$ absorbs the memory: $q^{-1-\varpi} \int_{q_0}^q s^\varpi \le (1+\varpi)^{-1}$.

Finite-resolution suprema have all moments uniformly in N; the terminal field is a distribution.

Regularity and the terminal field

Space. $K_q V_q = f_q$ with $\mathbb{E}\int_I \|f\|_2^2 \le C_I$ (energies), so $V \in L^2(I;H^1)$; $Y = V + X - \varrho(Y)$ has the regularity of the free field: H^σ, $\sigma < \frac{1}{2}$, in space; H^θ, $\theta < \frac{1}{2}$, in scale; tight in space–time L^p (and in $C(I;H)$: $C^{1/4}$ Hölder bound + uniform Fourier tails).

Terminal field. $H_q^2 \le \frac{C}{q} C_q$ gives, for $\frac{1}{2} < s < \frac{3}{2}$,

$$\int_Q^\infty \|(1-\Delta)^{-s/2} H_q\|_{\mathsf{HS}}^2 \mathrm{d}q \le C \int_Q^\infty \frac{1}{q} \sum_k \frac{(1+|k|^2)^{-s}}{q + c|k|^2} \mathrm{d}q \le C_s Q^{-(s-\frac{1}{2})},$$

and the drifts have squared tails $O(Q^{-3})$, $O(Q^{-5/2})$, $O(Q^{-3/2})$. Doob:

$$\sup_N \mathbb{E} \sup_{t \ge Q} \|Y_t - Y_Q\|_{H^{-s}}^2 \le C_s Q^{-(s-\frac{1}{2})}.$$

Corollary. *ν_N is tight in $H^{-s}(\mathbb{T}^3)$ for every $s > \frac{1}{2}$, with all moments uniformly in N. (Threshold of the free field in $d = 3$.)*

Identifying the limit

Lecture 6

The deterministic coefficients

Uniform bounds do not identify a limit: a weakly convergent vertex cannot in general be contracted against a varying covariance. Three tools:

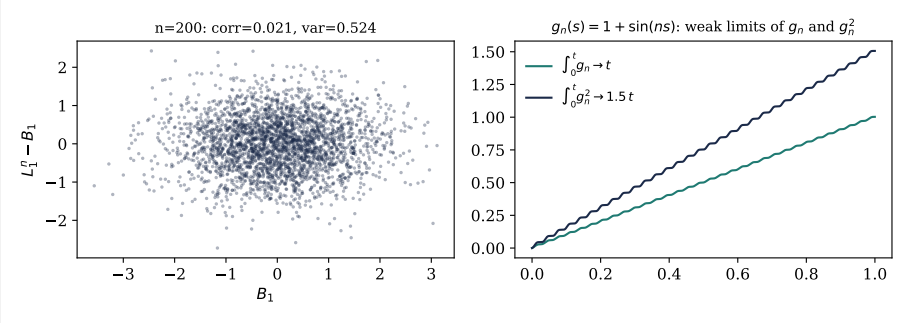
- **Weak vertex measures**: bounded rooted norms $\Rightarrow$ signed measures, weakly compact; the centred kernel in a first-moment topology. The limits solve the same fixed point, unique in its ball: the whole sequence converges.
- **Compact sandwich**: $G_N \to G$ in HS, $\mathsf{B}_N \to \mathsf{B}$ weakly, symmetric, uniformly bounded, $\sup_N \|G_N \mathsf{B}_N\| \le \theta < 1$, $\|G\mathsf{B}\| \le \theta \Rightarrow (I + G_N \mathsf{B}_N)^{-1} G_N \to (I + G\mathsf{B})^{-1} G$ in HS.
- **Contraction against a moving covariance**: rows of the inverses are Hölder + HS convergence $\Rightarrow (H^N)^2 \to H^2$ uniformly; weak vertices can be contracted.

Hence $H_q^N(Y_q^N) \to H_q(Y_q)$ in $L^2(I;\mathsf{HS})$ and $b_q^N(Y_q^N) \to b_q(Y_q)$ in $L^2(I;H)$.

Hidden noise

$L^n_t = \int_0^t g_n(s)\,\mathrm{d}B_s$, $g_n = 1 + \sin(ns)$: integrands $\to 1$ weakly, squares $\to \frac{3}{2}$.

$$(L^n, B) \Longrightarrow (B + 2^{-1/2}\tilde{B},\ B), \qquad \tilde{B} \perp B.$$



Left: $(B_1, L^n_1 - B_1)$, $n = 200$: correlation ≈ 0, variance of the gap $\approx \frac{1}{2}$ — the residual does not vanish and is orthogonal to B. Right: $\int_0^t g_n \to t$ but $\int_0^t g_n^2 \to \frac{3}{2}t$.

A limiting FBSDE obtained this way would carry an **extra martingale** orthogonal to the driving noise. We need **strong** compactness of the backward integrand $\Xi = \mathrm{Cov}\text{-}H(Y)$.

The Riccati equation of the posterior covariance

Lemma. $\mathrm{Cov}_q = D^2\,\mathscr{K}_q(P_q) \geq 0$ *and*

$$\mathrm{dCov}_q = -\mathrm{Cov}_q^2\,\mathrm{d}q + D^3\,\mathscr{K}_q(P_q)[\mathrm{d}W_q]$$

Lecture 3: conditional variance is resolved at rate $\mathrm{Cov}_t \dot{R}_t \mathrm{Cov}_t$; here $\partial_q R = 1$.

- $\langle f, \mathrm{Cov}_q f\rangle \geq 0$: a **nonnegative supermartingale**, initial value $\mathrm{Var}_{\nu_N}\langle \Phi_\infty, f\rangle$ bounded uniformly (uniform moments, Corollary).
- Stop at level K: decreasing part and bracket have bounded expectations ⇒ time regularity $W^{a,1}$, $a < \frac{1}{2}$ ⇒ strong compactness of each finite block.
- Spatial tails: isometry + symmetry, and terminal averaging $\psi_Q = \mathbb{E}_Q \int_L^{L+1} (Y_s + Z_s)\mathrm{d}s$ (H^σ bound on Y, $\mathbb{E}\|Z_s\|^2 \leq C s^{-5/2}$).

⇒ Cov^N tight in strong $L^2(I; \mathrm{HS})$; brackets $\int \mathrm{CovCov}^*$ and covariations with W converge.

Identification of the martingales

On a common filtered space (Skorokhod), along a subsequence:

- ψ is a continuous square-integrable martingale for the joint filtration; its bracket and its covariation with W are those of $\int \mathrm{Cov}\,\mathrm{d}W$;
- so $\psi - \int \mathrm{Cov}\,\mathrm{d}W$ has zero bracket and vanishes: **no extra martingale**;
- same for the forward equation, and $X = \int C\,\mathrm{d}W$ with the **same** W (needed to pass the thresholded energies).

$$Y_q = \int_{q_0}^{q} [b(Y) + H(Y)\,Z]\mathrm{d}s + \int_{q_0}^{q} H(Y)\,\mathrm{d}W,$$

$$Z_q = -\int_{q_0}^{q} [b(Y) + H(Y)\,Z]\mathrm{d}s + \int_{q_0}^{q} \Xi\,\mathrm{d}W, \qquad X_q = \int_{q_0}^{q} C_s\,\mathrm{d}W_s.$$

The result (I): convergence and the limiting FBSDE

Theorem. (subsequential construction) *For $\lambda > 0$, q_0 large, along a subsequence of lattices:*

a) (Y^N, Z^N, ψ^N, X^N) *converge in* $C(I;H)$, W^N *in* $C(I;H^{-r})$, $r > \frac{3}{2}$, (Cov^N, Ξ^N) *in strong* $L^2(I;\mathrm{HS})$, *jointly, a.s. on a new space;*

b) the limit solves the FBSDE on $[q_0, \infty)$ *with the continuum coefficients* b, H *(unique in their ball), an* H*-cylindrical Brownian motion obtained as the limit of the lattice innovations,* $\mathrm{Cov} = H(Y) + \Xi \geq 0$, *and* $\mathbb{E}\|Z_q\|^2 \leq C q^{-5/2}$;

ψ^N is tight in $C(I;H)$ because its brackets $\int \mathrm{Cov}^N \mathrm{Cov}^{N*}$ are tight and its Fourier tails are uniformly small.

The result (II): the terminal law

Theorem. (subsequential construction, continued)

a) $Y_q \to \Phi$ *in* H^{-s}, $s > \frac{1}{2}$, *a.s. and in mean square;* $\mathrm{Law}(\Phi) = \nu$ *is the weak limit of* ν_N, *and* $\langle Y_q + Z_q, h \rangle = \mathbb{E}[\langle \Phi, h \rangle \mid \mathcal{F}_q]$;

b) the limiting thresholded energies (built with $X = \int C \, \mathrm{d}W$*) obey the bounds of UV stability.*

The posterior structure survives: $\psi = Y + Z$ is a **closed observable**, Y is computed from the explicit action, Z is small at high resolution.

The renormalised field equation

Theorem. (integration by parts) $G(\varphi) = g(\langle \varphi, h_1 \rangle, \ldots, \langle \varphi, h_k \rangle)$, $g \in C_b^\infty$, h, h_i *smooth:*

$$\lim_N \mathbb{E}\left[G\left(\iota_N \Phi_\infty^N\right) \langle \mu_N \Phi_\infty^N + \lambda (\Phi_\infty^N)^3, j_N h \rangle \right] = \mathbb{E}_\nu \left[D_h G(\Phi) - G(\Phi) \langle \Phi, (q_0 - \Delta)\, h \rangle \right].$$

Proof: lattice Gaussian integration by parts in $\mathrm{e}^{-V_N - \frac{1}{2}\langle \varphi, K_{q_0} \varphi \rangle}$; $K_{q_0}\, j_N\, h = j_N\, (q_0 - \Delta_N^{\mathrm{tr}})\, h$; tightness and uniform moments.

- The terminal case of the DS equations of Lecture 3.
- The renormalised cube $\mu_N \Phi + \lambda \Phi^3$ is identified only **tested in expectation**, not as a random distribution.
- Consistent with the expectation (Lecture 3) that the limiting force is an observable without terminal representative; this is **not proved**.
- At every finite resolution the force is explicit: $F_q = K_q Z_q - D \bar{\Gamma}_q(Y_q)$.

What is not proved

- **Uniqueness** of the limit, of the FBSDE, independence of the regularisation.
- **Small mass / strong coupling**: the construction needs $\gamma\, q_0^{-1/2}$ small, i.e. weak coupling $\lambda/m \ll 1$. Removing it needs a priori estimates without the small feedback γ^2/q_0, and a coefficient construction outside the perturbative ball.
- **Axioms**: reflection positivity holds on the lattice (lattice and mid-lattice planes) and passes to ν for every coordinate reflection (standard; exercise); rotation invariance and clustering need infinite volume.
- **Infinite volume**: the force and its Burgers flow are intensive; estimates must be made local. Uniform-in-volume coefficient bounds and local energy moments are open; retaining the full FBSDE also needs control of the **noise from infinity** in the nonlocal Ξ.

Summary of the course

Wightman → Euclidean (L1) → free field (L2) → scale decomposition: Polchinski, FBSDE, variational problem (L3) → Φ^4_3 through the FBSDE in forecast coordinates (L4–6).

Lecture 3 (formal)	Lectures 4–6	in the limit
forecast $\psi_t = \mathbb{E}_t \Phi_T$	$\psi = Y + Z$	closed martingale
response $H_t = \mathrm{Cov}_t$	$\mathrm{Cov} = H(Y) + \Xi$	strong $L^2(\mathrm{HS})$, Riccati
Γ_t, Wetterich	$\bar{\Gamma}_q$ explicit, defect e	deterministic coefficients
force F_t	$K_q Z_q - D\bar{\Gamma}_q(Y_q)$	observable at finite q
DS equation	integration by parts	tested in expectation

UV stability = an explicit action predicts the conditional mean up to an error that is small at high resolution.